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D³CNet: Long-term time series prediction based on dual decomposition and dual-channel hybrid network

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ARTICLE INFO

Keywords:

Long-term time series forecasting
1D-CNN and GRU
Dual-channel hybrid network
Mode decomposition

ABSTRACT

Recently, time series forecasting has been widely used in many fields. Time series feature extraction is a key research area, and the complexity and diversity of data make traditional feature decomposition methods difficult to adapt. Aiming at the problem that it is difficult to decompose the mixture of different scale information, and the single-channel model cannot capture the long-term correlation and local features in the multivariate time series at the same time, this paper proposes a long time series forecasting method based on dual decomposition and dual-channel hybrid network (D³CNet). The method utilizes S-EMD to decompose time series data at multiple scales, with each channel in the hybrid network used to process different aspects of the data. This method can capture both global and local information to make more accurate predictions. First, the EMD is utilized to decompose the time series data into components of different frequencies. Then, the data is seasonally decomposed to obtain smoother time series information and reconstruct the seasonal components. Finally, the data are fed into a D³CNet model to predict time series target values. In this paper, multivariate and univariate experiments are conducted on six commonly used datasets. The average MSE and MAE of D³CNet proposed in this paper are significantly improved on the datasets. In the ETT dataset, D³CNet reduces the average MSE by 29.4 % and MAE by 15.7 %. This further proves that D³CNet can reduce the error and improve the accuracy, thus also proving the effectiveness and superiority of the method.

1. Introduction

Time series forecasting is usually based on historical observations and is used to predict the future data values of a certain attribute in a specific field. The role of time series prediction lies in its ability to reveal inherent patterns in data, such as trend, seasonality and periodicity, which are of great significance for people's decision-making, planning, and risk management. It occupies a critical position within scientific and engineering fields. Specific applications span financial stock research (Gong and Xing, 2024), traffic flow analysis (Kong et al., 2025), disease transmission investigation (Matsubara et al., 2014), and weather change prediction (He et al., 2025), and more. For instance, enhanced precision in traffic flow prediction can empower traffic management authorities to more effectively oversee and regulate road traffic states, mitigate congestion, and augment the utilization efficiency of road infrastructure (Qin et al., 2021). Accurate weather forecasts also play an important role in people's planning and activity scheduling in life, which can help people avoid the inconvenience and danger caused by bad weather (Zhao et al., 2021). With the help of accurate inventory forecasts,

investors can develop smarter strategies to cope with changing market conditions and reduce the risk of inventory backlogs and stock-outs (Ma et al., 2022, Teng et al., 2025). Accurate electricity forecasting is the lifeline that enables the smooth functioning of electricity markets and energy trading, helping to balance supply and demand, reduce energy costs, and support the integration of renewable energy sources (Lai et al., 2018, Ma et al., 2024).

In early studies, most scholars at home and abroad generally relied on traditional statistical methods to forecast time series data. Wang Y. et al (Yingwei and Shucai, 2021) used ARIMA model for time series forecasting, Faghih S. (Faghih et al., 2020) used ARMA and linear regression for modelling and calculating cab demand. Although these traditional statistics-based methods show their unique advantages in dealing with linear relationships, their data requirements are also an obvious limitation of the method. Subsequently, the emergence of machine learning methods can well improve the shortcomings of traditional statistical methods, such as Feng, et al (Feng and Niu, 2021) artificial neural networks (ANNs) were utilized to perform predictive analysis on river flow safety, and Singh, et al (Sasal et al., 2022) used the SVM model and

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Received 16 April 2025; Received in revised form 25 November 2025; Accepted 22 March 2026

Available online 24 March 2026

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repeated wavelet transform to predict the electrical load. However, since machine learning usually treats data as independent of each other, it is unable to capture the dependencies between data as well as their seasonality and periodicity when predicting complex time series, resulting in no significant improvement in the effect. Because, natural language processing is still temporal in nature and deep learning has achieved remarkable success in natural language processing tasks, these deep learning methods widely used in natural language processing have also been introduced to time series research. for example, Jian Liu (Liu et al., 2023) et al. used a Long Short-Term Memory (LSTM) network for hourly step-by-step prediction of solar irradiance, and Bin Yang et al. (Yang et al., 2023) used a Gated Recurrent Unit (GRU) network to predict the motion of a non-stop cardiac surgery. The model contains only two gated structures and has comparable performance to LSTM with all hyperparameters tuned, but the GRU structure is simpler and easy to implement with few training samples. However, during long - sequence prediction tasks, the model may encounter challenges such as gradient explosion or gradient disappearance.

Some research methods decompose a time series into several different subseries for separate processing. Decomposition methods (Chaovalit et al., 2011) are able to decompose a time series into a number of subseries of different sizes. These sub-series tend to exhibit more pronounced smoothness and regularity compared to the original series (as shown in Fig. 1), making it easier for deep learning models to mine potential features. A multitude of research studies have indicated that performing a decomposition of time - series data to individually analyse and construct models for the seasonal and trend components can enhance the predictive performance (Raghuvanshi, 2024). For example, the SCINet (Liu et al., 2022) extracts valuable temporal features from these sequences or features by recursively down sampling time series into subsequence using multiple gradients. By combining the decomposed multiple sequences to enrich the original sequence features, SCINet can effectively forecast time series with complex temporal dynamics. Autoformer (Wu et al., 2021) uses a moving average-based time series segmentation strategy to partition time series data into a trend-periodic portion and a seasonal portion, which is suitable for progressively extracting long-term stable trends from predicted hidden intermediate indicators. The empirical modal decomposition (EMD) is used to signalize the time series into intrinsic modal functions (IMFs), each of which corresponds to a different frequency component and trend information, which helps to reveal underlying patterns in the data, such as long-term trends, seasonal variations, and cyclical fluctuations. In theory, EMD decomposition techniques aim to reveal the trend, seasonal and cyclical components in time series data. However, in practice, the problem of modal mixing is often encountered. That is to say, signal components of different frequencies are interleaved to form a single IMF. This phenomenon affects the accuracy of the decomposition results and makes it unreliable to directly utilize the EMD decomposition results. In addition, for the analysis of multi-scale subsequence existing studies tend to merge the decomposed subsequence for processing to consider their interrelationships, a practice that may weaken the correlation between the sequences. To address the problem of modal aliasing in existing decomposition methods and the failure of the trend and seasonal components of the subseries to take full advantage, we have

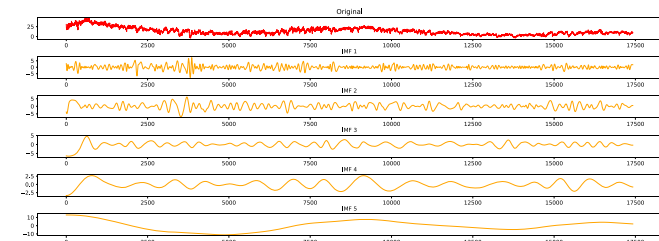


Fig. 1. Schematic diagram of intrinsic rules in time series.

optimized the decomposition process to reduce the modal aliasing phenomenon and make effective use of the multiscale information.

In the process of modeling time series, long time series tend to exhibit patterns of complexity relationships, such as periodicity and trend. So far, many scholars used Encoder module and Decoder module in Transformer (Vaswani, 2017) model. Transformer model has more parameters, so it is limited by the computational complexity, and the original Transformer is not good at modeling prediction for long time series. Informer (Zhou et al., 2021) improved the Transformer by reducing the complexity of the model to improve the Transformer, which led to the derivation of a new Transform architecture. Subsequently, several Transformer variants have been proposed in the field of time series forecasting (Ilbert et al., 2024, Liu et al., 2023), which are mainly improved by increasing the performance or computational efficiency of the model. Recently, the simple and basic linear layers have been increasingly used for trend analysis (Wang et al., 2023, Zhao et al., 2024). However, it ignores the potential complex structural relationships and correlations of data from different fields. On the contrary, Convolutional Neural Networks (CNNs) have the ability to efficiently extract local features and temporal trends from time series data. In addition, compared with Transformer based models, CNN based models have the characteristic of reduced time complexity. So far, the improved CNN models have achieved further improvements in various fields. For example, they have achieved good results in audio recognition, real-time traffic vehicle prediction, and financial stock price prediction (Liu et al., 2022), which simplify the computational process to a certain extent, but are weaker in capturing long-term features. Therefore, if we are going to use CNN to process time series data, we should not only focus on the trendiness and local features, but also fully extract the time dependence of the data, which can help to improve the model prediction accuracy and reduce the error with the real value.

In response to the previous decomposition methods, there is a mixture of information at different scales, and the prediction methods often do not fully utilize the information and it is difficult to explore the intrinsic correlation between the data, thus restricting the improvement of the prediction accuracy. Therefore, based on dual decomposition and dual-channel hybrid networks, a long time series prediction method is proposed (D³CNet). In order for the model to maximize the use of time series features, D³CNet employs different analysis strategies for different scales of information. After the EMD, in order to separate the information more thoroughly and improve the smoothness of the subsequence, we seasonally encode the subsequence and reconstruct the seasonal components. For time series prediction, we used a two-channel hybrid network to improve the performance of the prediction model. On the one hand, the convolutional neural network (CNN) is used for in-depth local feature extraction and trend analysis of the decomposed subsequence on the other hand, the gated recurrent unit (GRU) is responsible for processing the raw data and is used to obtain the long-term dependencies in the sequences, thus overcoming the limitations of CNNs in processing long time series data. By using a dual-channel hybrid network, the model is able to efficiently process data at different time scales. Among them, the CNN channel focuses on capturing short-term patterns, while the GRU channel focuses on analysing long-term trends. This design fully utilizes the advantages of each model and integrates information from multiple channels, improving the accuracy of the prediction results. We demonstrate the effectiveness of D³CNet through extensive experiments.

The major contributions of this paper are as follows:

- For long time series prediction, we construct a dual-channel hybrid network model based on decomposition, which realizes more efficient and accurate prediction.
- We propose an S-EMD module for effective decomposition of complex time series. The model adds seasonal factors on the basis of EMD, which can extract multi-scale features more effectively. And S-EMD makes the time series data smoother by adding seasonal

components, optimizes the iterative process, and thus improves the overall computational efficiency.

- We designed a dual-channel hybrid network in which GRU and CNN process the raw and decomposed time series data, respectively. The dual-channel hybrid network allows each channel to focus on a different aspect of the data features, which enhances the robustness of the model by integrating the prediction results from different models and also reduces the possibility of overfitting. The dual-channel hybrid network combines the strengths of different models to enhance the model's comprehension of complex time series data and improve the accuracy of model predictions.

The remaining content of this article is distributed as follows: In [Section 2](#), existing literature on time series prediction and the limitations of related methods are listed. [Section 3](#) presents the methodology used in this paper as well as related techniques. [Section 4](#) presents the dataset. And the analysis of experimental results. Section 5 summarizes the work of this paper and the outlook for the future.

2. Related work

To improve the accuracy and predictive range of time series forecasting, we need to dig out as much as possible the data characteristics and intrinsic relationship of the data when making forecasts. In recent years, extracting relevant features from these complex time series and realizing high-precision forecasts have become more challenging as the complexity and volume of data increase.

In traditional time series prediction methods, there are simple and highly interpretable models such as AR, ARMA, and Prophet in statistics ([Taylor and Letham, 2018](#)). However, these models tend to rely on smoothness assumptions for the data, which limits their ability to be applied in dealing with today's large-scale data. Their inability to effectively handle the complexity of multidimensional data and their neglect of pattern recognition at different scales are limitations that reduce the utility of these traditional approaches in real-world complex forecasting tasks.

Deep learning-based methods have also been widely used in time series forecasting. For example, TRNN, based on the RNN method, achieved a significant increase in prediction accuracy by improving the neural network architecture and optimizing the prediction window. The problems of gradient vanishing and gradient explosion have been present in RNNs, and some new research work has been proposed to be able to model complex time series. These research works include the use of structures such as LSTM and GRU to enhance the ability of RNNs to better handle long sequence data and avoid the gradient problem. In addition, other researchers have also explored RNN models that combine attention mechanisms. Zhu et al. ([Zhu et al., 2019](#)) fused LSTM with an encoder decoder framework based on attention mechanisms for workload prediction in large-scale operations in data centers. This method significantly improves the accuracy of workload prediction by accurately capturing correlations in the data. In order to explore the correlation between the distance of historical data points and the current time window, Song et al. ([Song et al., 2018](#)) used an LSTM model with an adaptive structure. The dataset was divided into immediate and cumulative information, and optimized LSTM units were subsequently used to replace the hidden layer in traditional RNNs. This approach was taken to more efficiently obtain long-term dependence features in time-series data. In order to better model and predict multi-scale dynamics, DTS-ESN ([Tanaka et al., 2022](#)) placed distributed leakage rates in the neurons of the storage layer. Despite its capacity to capture multiscale information, the model's expressiveness is constrained when handling large-scale and high-dimensional data. Additionally, it exhibits significant sensitivity to parameter selection, and inappropriate parameter settings may lead to degradation of model performance.

Transformer-based models, relying on their self-attention mechanism, have the advantage of excelling in extracting long-term

dependency information in the features of time-series data and maintaining the relative completeness of the information over a wider range of series. Consequently, Transformer has emerged as a prevalent technique for time series (TS) forecasting, and many variant models of Transformer have been proposed for time series forecasting. For example, FEDformer ([Zhou et al., 2022](#)) by introducing Fourier and wavelet transforms in encoders and decoders; In fact, it replaces the self-attention and cross attention mechanisms in the encoder and decoder with Fourier transform and wavelet transform, while Pyraformer ([Liu et al., 2022](#)) built transient models on different scales. However, these models are characterized by high complexity and difficulty in capturing temporal dynamics. The self-attention mechanism limits the performance of Transformer and its derivative models in long-term prediction, as it introduces quadratic time complexity when calculating weights. In the field of time series forecasting, MLP has also been introduced. Recent studies have found that the underlying linear models can also show excellent prediction results in some specific scenarios. For example, The N-BEATS framework ([Oreshkin et al., 2019](#)) is a model based on forward backward residual networks, which provides good interpretability. LTSF Linear ([Zeng et al., 2023](#)) is a single linear model that performs better than basic Transformer models in certain fields. DLinear ([Zhao et al., 2024](#)) uses only two single-layer linear layers of the network for prediction, which reduces complexity. However, it lacks sufficient model complexity to learn these complex time series features and is prone to overfitting when the model is oversimplified.

Some popular and recent forecasting architectures then incorporate time series decomposition techniques to enhance the ability to model time series patterns. Decomposing time series is a common method for analyzing complex time series data, which usually splits the time series into three main components: trend information, seasonal component and random impact information. The trend information reveals the long-term evolution direction of the time series, the seasonal component reflects the cyclical fluctuations of the time series, and the random effects information represents the abnormal data fluctuations or data noise in the data. Wavelet decomposition, as an effective tool for multi-scale analysis, can clearly distinguish different frequency components and is widely used in the field of time series prediction, helping models accurately capture complex time series characteristics. Veg-W2TCN ([Rhif et al., 2023](#)) fused wavelet transform and temporal convolutional network (TCN) to construct an integrated forecasting framework for estimating vegetation cover. Wave-Former ([Wei et al., 2025](#)) was developed by applying maximal overlap discrete wavelet transform (MODWT) to realize multi-period decomposition of time series, and designing a novel wavelet attention (WA) mechanism to specifically extract periodic features of different frequencies while weakening the interference of outliers. Xia Jin et al. ([Jin et al., 2023](#)) used empirical modal decomposition and discrete Fourier transform to obtain the trend component and period component to achieve the prediction of short-term traffic flow. However, most studies are too limited in scale selection, often focusing only on a single scale or randomly selecting scales for analysis. This greatly limits the comprehensive analysis of complex patterns, making it impossible to fully capture the complex relationships in time series data and utilize information from different scales.

Empirical modal decomposition, as a multiscale feature extraction method, the time series is decomposed into a series of inherent modal functions (IMFs), where each IMF captures the characteristics of the signal at a specific time scale. These IMFs contain local features of signals and multi-scale information of signals, making it an effective tool for trend analysis, which is particularly suitable for application to the analysis of multivariate time series.

However, extant studies have focused on specific aspects of the data, modeling based on the characteristics of a certain scale, or applying a unified training model to different scales. This approach may neglect certain scale information when predicting multivariate time series, thus affecting the accuracy of prediction. Therefore, it is particularly

important to design a dual-channel hybrid network to model multiscale features. This network structure can deal with different variables separately, so that the model can understand the degree of influence and dependence correlation between the features, which can achieve the purpose of reducing prediction error and improving accuracy.

3. Method

In this section, we introduce the techniques and methods used in the D³CNet proposed in this paper. Finally, a detailed overview of the model components is provided.

Since time series data exhibit characteristics such as seasonality, trend, periodicity, etc., and are decomposed into subseries containing different frequency components after time series decomposition. Therefore, in order to fully utilize the characteristics of time series, we propose a quadratic decomposition and dual-channel hybrid network time series prediction method to learn the complex patterns in the long time series. As illustrated in Fig. 2, we integrate the empirical modal decomposition (EMD-Block) method, which is a hallmark of signal processing. This approach does not necessitate a predefined basis function, instead decomposing the signal according to its intrinsic local features while preserving the signal's original characteristics. Consequently, the time series is fragmented into multiple intrinsic modal functions (IMFs). A detailed analysis of these IMFs is then conducted to identify those that contain seasonal patterns. These seasonal IMFs are extracted and reconstructed into seasonal components (SC-Block). Subsequently, the original data and the data after secondary decomposition were input into a dual channel hybrid network. A convolutional layer and pooling layer were added to the first channel to extract features, and two GRU layers were added to the second channel to further process the dependencies between sequences. Finally, we merged the features of the two channels using a concatenate layer for information interaction, and added a fully connected layer and an output layer to complete the final prediction.

3.1. Empirical modal decomposition module

The decomposition process of EMD requires the signal's own temporal attributes, so it is adaptive. The EMD decomposition of IMFs subsequences can be summed up to obtain the original signal, which provides a more accurate regularity representation of the complex signal data, and provides a complete framework for signal analysis, thus enhancing its modeling capability, which is conducive to time series modeling. As shown in the EMD-Block section of Fig. 2, we used the EMD

module to decompose the time series into a series of intrinsic modal functions (IMFs). This iterative method can effectively decompose complex time series signals and realize efficient decomposition and adaptive segmentation of complex signals. Specifically, the EMD module works by extracting the first IMF based on the local characteristics of the signal by identifying the extreme points of the signal and constructing the upper and lower envelopes, and then calculating the average of these envelopes. This process is iterated until all IMFs have been extracted or until the remaining signals have become sufficiently smooth to be decomposed into new IMFs. In this way, any complex signal can be viewed as the superposition of the sum of multiple intrinsic modal functions of different frequencies and amplitudes. For example, the time series of the ETTh1 dataset after EMD decomposition is shown in Fig. 3.

The following steps describe the detailed process of EMD decomposing time-series data:

Step 1:

Input the original signal $x(t)$, identify all the extreme values, and connect the local extremely large value points into an upper envelopes and local minima into a lower envelope by means of a cubic spline curve. All data points are between the upper and lower envelope ranges.

Step 2:

The mean value $m_1(t)$ of the upper and lower envelopes is calculated. Eq. (1) is obtained:

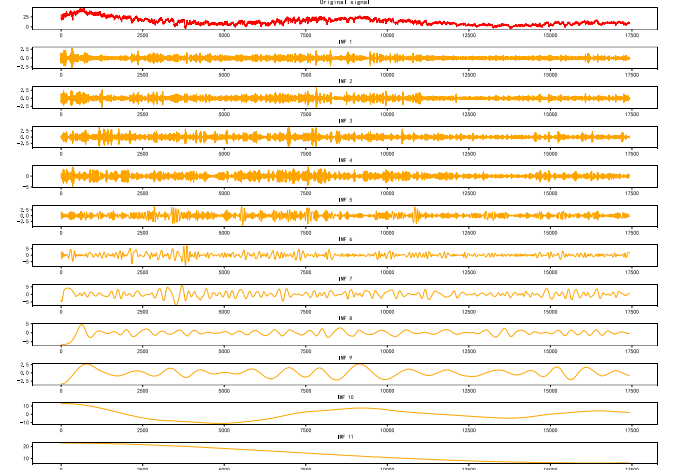


Fig. 3. Decomposition of the time series of ETTh1.

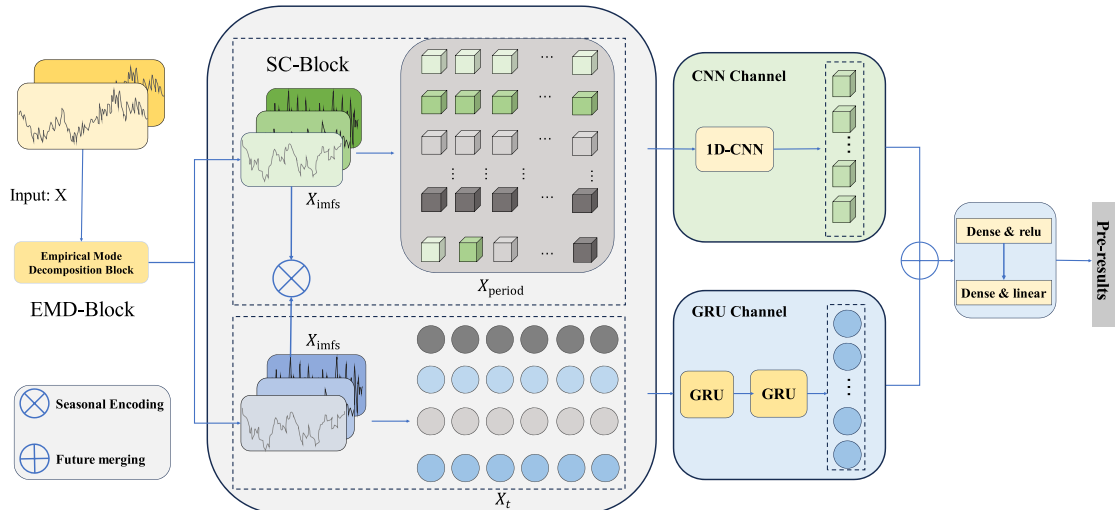


Fig. 2. General diagram of the model of D³CNet.

$$h_1(t) = x(t) - m_1(t) \quad (1)$$

In the event that $h_1(t)$ fulfills the condition of IMF, it may be designated as the initial IMF component of $x(t)$.

Step 3:

In the event that $h_1(t)$ fails to meet the condition of IMF, $h_1(t)$ is considered the original data. Steps 1 and 2 are reiterated to ascertain the mean of the upper and lower envelopes, $m_1(t)$. The mean is subsequently calculated to arrive at Eq. (2):

$$h_1(t) = h_1(t) - m_1(t) \quad (2)$$

See whether $h_1(t)$ satisfies the necessary conditions for the IMF component, if not, repeat the two steps as above k times until the premise is satisfied to obtain. the two criteria that must be satisfied by the IMF are as follows:

- (1) The main criterion for the Initial Modal Function (IMF) is to specify that the number of times over zero and extreme points should be the same or differ by no more than one throughout the time series.
- (2) The IMF requires that the mean of the envelope be zero at every point throughout the dataset and by continuously performing the filtering process until Eq. (3) is satisfied.

$$h_{1k}(t) = h_{1(k-1)}(t) - m_{1(k-1)}(t) \quad (3)$$

If $h_{1k}(t)$ satisfies the above conditions, the first IMF component is then represented by Eq. (4):

$$c_1(t) = h_{1k}(t) \quad (4)$$

Step 4:

Separate $c_1(t)$ from the signal $x(t)$ to obtain $r_1(t)$ as in Eq. (5):

$$r_1(t) = x(t) - c_1(t) \quad (5)$$

Repeat the above three steps with $r_1(t)$ as the original signal for n cycles to obtain the next IMF component $c_2(t)$ until the n th IMF component.

Step 5:

The sum of all our IMF components and residual components is the original signal $x(t)$. When $r_n(t)$ becomes a monotone increasing function or a monotone decreasing function, the other $r_n(t)$ becomes a residual component. The Eq. of the original signal $x(t)$ is Eq. (6):

$$x(t) = \sum_n c_i(t) + r_n(t) \quad (6)$$

The method is executed iteratively. After that, n iterations are performed and the process produces the n th order IMF component $c_n(t)$ and the corresponding residual $r_n(t)$, when the residual $r_n(t)$ exhibits either a constant tendency or a monotonic trend.

Step 6:

Use correlation analysis to filter out high-frequency noise data following EMD decomposition.

$$r_{x,c_i} = \frac{\sum_{t=1}^N [x(t) - \bar{x}][c_i(t) - \bar{c}_i]}{\sqrt{\sum_{t=1}^N [x(t) - \bar{x}]^2} \cdot \sqrt{\sum_{t=1}^N [c_i(t) - \bar{c}_i]^2}} \quad (7)$$

$c_i(t)$ is the mean of the original signal $x(t)$, $\bar{c}_i$ is the mean of the i th IMF $c_i(t)$, and r_{x,c_i} is the correlation coefficient between $x(t)$ and $c_i(t)$.

Step 7:

Subtract the noise signal from the sequence obtained after EMD decomposition.

$$Y_t = x(t) - c_p(t) \quad (8)$$

$$Y_t = T_t \times S_t \times E_t \quad (9)$$

$c_p(t)$ represents high-frequency noise, while Y_t denotes the sequence

after EMD decomposition and denoising. The entire EMD decomposition flowchart is illustrated in Fig. 4.

3.2. Reconstructing the seasonal component module

Empirical Mode Decomposition (EMD) of time series data may result in mode overlap, complicating the accurate extraction of seasonal components—a critical step for long-term time series forecasting. Reconstructing seasonal components enables more precise capture of periodicity, highlights features, and reveals residual nonlinear characteristics. Therefore, a reconstructed seasonal component module was designed to enhance data feature extraction, as shown in the SC-Block section of Fig. 2. During seasonal decomposition, to enhance the independence of each component for easier interpretation and analysis, we transformed the multiplicative model into an additive model. The flowchart for reconstructing the seasonal component is shown in Fig. 5, with detailed steps as follows:

Step 1: Logarithmic Transformation

Perform a logarithmic transformation on the sequence Y_t obtained after EMD denoising. This converts $Y_t = T_t \times S_t \times E_t$ into an additive relationship, corresponding to formula (10):

$$\log(Y_t) = \log(T_t) + \log(S_t) + \log(E_t) \quad (10)$$

Among these, Y_t is the time series after EMD decomposition and denoising, T_t is the trend information, S_t is the seasonal data, and E_t is the random residual value.

By converting the coupling amplification effect between trend and seasonal components in the multiplicative model into a linear superposition relationship through logarithmic transformation, this approach effectively mitigates estimation errors in seasonal components caused by sharp fluctuations in the trend. Not only does it eliminate interference from heteroscedasticity in the original series, enhancing the robustness of decomposition results, but it also paves the way for precise extraction of trend components using moving average (MA) techniques in subsequent steps.

Step 2: Estimating Trends Using Moving Averages

Based on the log-transformed sequence $\log(Y_t)$, a moving average (MA) is employed to estimate the trend component $\log(\widehat{T}_t)$ within the sequence $\log(Y_t)$, as shown in formula (11).

$$\log(\widehat{T}_t) = \frac{1}{k} \sum_{i=0}^{k-1} \log(Y_{t-i}) \quad (11)$$

Among these, $\log(\widehat{T}_t)$ represents the estimated value of the trend component at time point t , and k denotes the size of the moving average window. The window size for moving averages is typically set equal to the length of the seasonal cycle, so the moving average window size k is defined as equal to the seasonal cycle length m . For even-length cycles, a two-step moving average method may be required to correctly synchronize the moving average, ensuring the trend component aligns with the original sequence's time index.

Step 3: Trend Removal and Seasonal Component Extraction

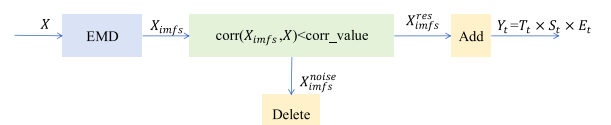


Fig. 4. Decomposition flowchart of EMD.

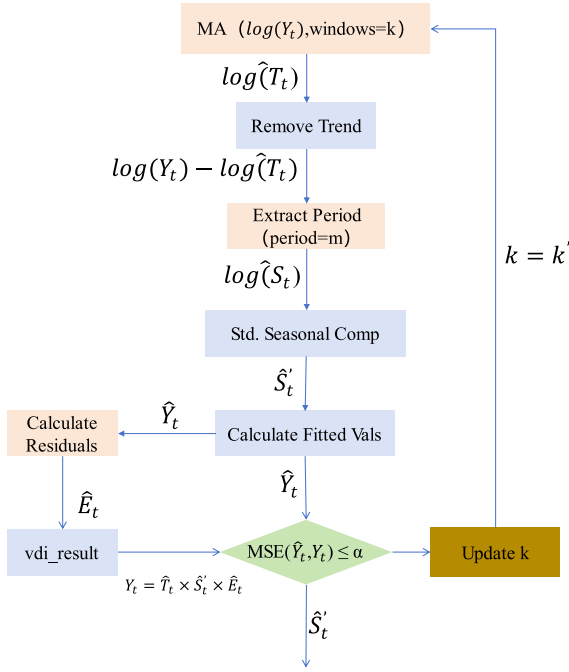


Fig. 5. Seasonal reconstruction flowchart.

(1) Subtract the estimated trend component $\log(\hat{T}_t)$ from the log-transformed sequence $\log(Y_t)$ to obtain the trend-removed sequence, as shown in Eq. (12).

$$\log(Y_t) - \log(\hat{T}_t) \quad (12)$$

This sequence is essentially $\log(S_t) + \log(E_t)$, containing only the superimposed information of seasonal and residual components.

(2) Group the detrended sequence $\log(Y_t) - \log(\hat{T}_t)$ according to seasonal cycles m . Calculate the mean of each group as the logarithmic estimate of the seasonal component $\log(\hat{S}_t)$, as shown in formula (13):

$$\log(\hat{S}_t) = \frac{1}{m} \sum_{j=0}^{m-1} (\log(Y_{t-i}) - \log(\hat{T}_{t-j})) \quad (13)$$

Among these, $\log(\hat{S}_t)$ is the estimated value of the seasonal component at time point t , and m is the length of the seasonal cycle.

Step 4: Seasonal Quantity Standardization

In the multiplicative model, to ensure the symmetry of seasonal fluctuations within the cycle, the mean of the seasonal component S_t must be 1.0. Therefore, $\log(\hat{S}_t)$ must first be converted back to the original scale via formula (14) using an exponential transformation, followed by standardization according to formula (15).

$$\hat{S}_t = e^{\log(\hat{S}_t)} \quad (14)$$

$$\hat{S}'_t = \frac{\hat{S}_t}{\frac{1}{m} \sum_{j=0}^{m-1} \hat{S}_j} \quad (15)$$

After standardization, mathematical derivation yields $\frac{1}{m} \sum_{j=0}^{m-1} \hat{S}'_j =$

1, satisfying the mathematical constraint for seasonal components in the multiplicative model. This approach effectively mitigates interference in subsequent forecasts caused by uneven seasonal fluctuation magnitudes within the cycle.

Step 5: Inverse Logarithmic Transformation and Residual Calculation
(1) Calculation of Fitted Values: Substitute the detrended $\log(Y_t) - \log(\hat{T}_t)$ and the standardized seasonal component $\log(\hat{S}'_t)$ back into the Eq.. Through inverse logarithmic transformation, restore the data to its original scale to compute the fitted values of the sequence, as shown in formula (16):

$$\hat{Y}_t = e^{\log(Y_t) - \log(\hat{T}_t) + \log(\hat{S}'_t)} = e^{\log\left(\frac{Y_t \times \hat{S}'_t}{\hat{T}_t}\right)} = \frac{Y_t \times \hat{S}'_t}{\hat{T}_t} \quad (16)$$

Among these, $\hat{T}_t = e^{\log(\hat{T}_t)}$ represents the raw scale estimate of the trend component.

(2) Residual calculation: Estimate residual components by measuring the deviation between fitted values and the original sequence.

$$\hat{E}_t = \frac{Y_t}{\hat{T}_t \times \hat{S}'_t} \quad (17)$$

This residual satisfies the multiplicative model $Y_t = \hat{T}_t \times \hat{S}'_t \times \hat{E}_t$, with a mean close to 1 and no significant periodicity, verifying the completeness of the decomposition.

Step 6: Model Accuracy Validation

Evaluate the accuracy of the decomposition by comparing the original data Y_t with the fitted values $\hat{Y}_t$ and using metrics such as Mean Squared Error (MSE) or Mean Absolute Error (MAE).

$$MSE = \frac{1}{n} \sum_{t=1}^n (Y_t - \hat{Y}_t)^2 \quad (18)$$

$$MAE = \frac{1}{n} \sum_{t=1}^n |Y_t - \hat{Y}_t| \quad (19)$$

The above formulas constitute the basic framework of the seasonal decomposition additive model. Through the seasonal decomposition process, we extract smoother time series features with seasonality. Subsequently, we normalize the time series data and the raw data after S-EMD decomposition, and then input them into our dual-channel model architecture for prediction respectively. The specific decomposition process performed by S-EMD is shown in Fig. 6.

Pseudo code for feature extraction module using S-EMD decomposition:

Algorithm: S-EMD-based feature extraction.

3.3. Dual channel hybrid network

The decomposed multi-scale information cannot be fully utilized, which makes the local features as well as trends of the data insufficiently captured, thus increasing the error of model prediction. A two-channel hybrid network is designed, which consists of a GRU-based channel dedicated to processing raw time series data to identify long-term dependencies in the data, and a CNN-based channel responsible for processing decomposed time series data. This design allows the model to focus on both global and local features of the time series data, thus

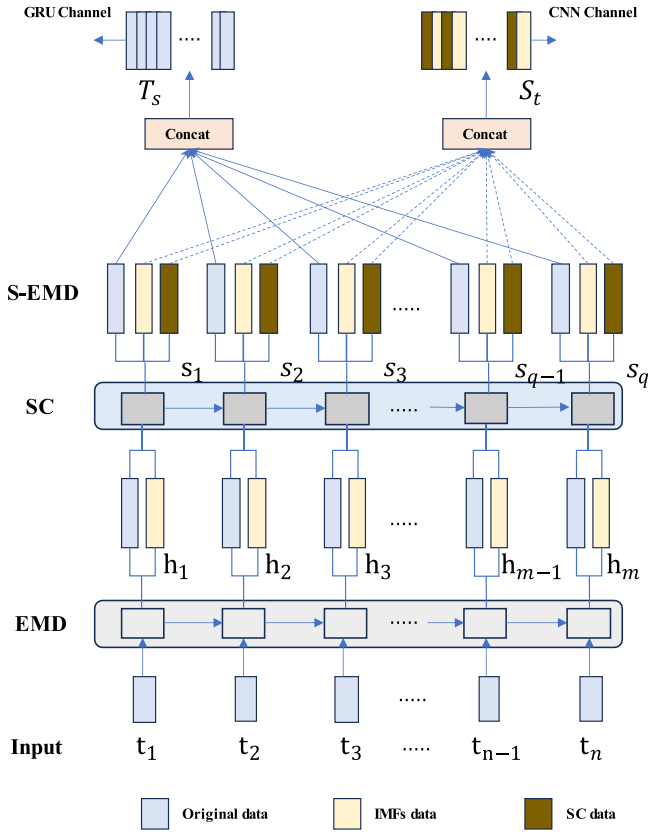


Fig. 6. Feature extraction process of S-EMD.

improving the prediction accuracy. The performance of the model is further enhanced by effectively integrating the features extracted from the two channels. The structure of this dual-channel hybrid network is shown in Fig. 7 below.

3.3.1. Inputs for dual-channel hybrid network

Time series data have different scales of input values, and many algorithms and models are sensitive to the scale of the input data, which may reduce the convergence speed and performance of the model. So, we normalized and counter-normalized the data. The standardization formula is shown in (20):

$$Z = \frac{x - u}{\sigma} \quad (20)$$

where x is the original data, u is the mean, and σ is the standard deviation.

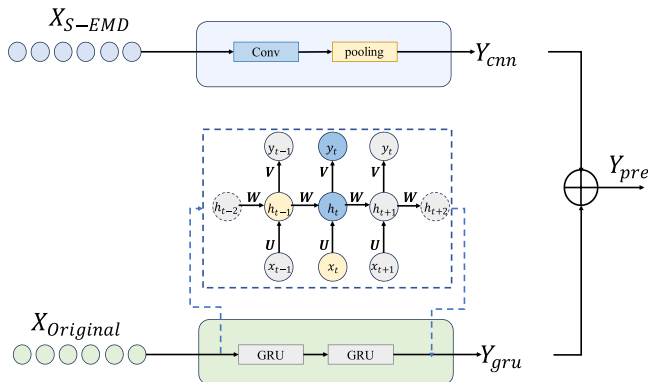


Fig. 7. Dual channel hybrid network.

The inverse normalization Eq. is as in (21):

$$x = Z \cdot \sigma + u \quad (21)$$

Where σ is the standard deviation of the original data, u is the mean of the original data, and Z is the standardized data. This standardization and back-standardization process ensures the consistency and comparability of the data in the pre-processing and post-processing stages.

3.3.2. Dual-channel network

The GRU channel and CNN channel are used for feature extraction after normalization of the original and secondary decomposed data. The GRU channel focuses on identifying long-term dependencies in the data, while the CNN channel focuses more on preserving local and seasonal features.

GRU is used to solve the problem of gradient vanishing in RNNs. GRU controls the flow of information by introducing the Update Gate and Reset Gate mechanisms, which can be realized to identify and preserve important information, while ignoring irrelevant or redundant components, and ensure that the information is effectively passed between different time steps, thus effectively dealing with the long-term dependency issues. The memory in GRU focuses on capturing long-term dependencies in the time series, while the CNN channel is more concerned with capturing local features and seasonal features. The memory unit in the GRU facilitates the network's ability to capture significant information in the sequence by storing the current state and updating it to a new state, thereby enhancing the network's capacity to address long dependencies.

The input sequence of $X = [x_1, x_2, \dots, x_t]$, x_t is the input of the t th time step. the extraction process of the long-term dependency of GRU is as follows:

The hidden state at the previous time step is h_{t-1} , the current candidate hidden state is $\tilde{h}_t$, and the reset gate r_t represents the effect of h_{t-1} on $\tilde{h}_t$. As in Eq. (22), σ is the sigmoid function and σ is the trainable weight matrix.

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r) \quad (22)$$

r_t represents the reset gate used to control the previous time step h_{t-1} 's effect on the hidden state, the formula for the former candidate hidden state $\tilde{h}_t$ is given in (23).

$$\tilde{h}_t = \tanh(W_h \cdot [r_t \odot h_{t-1}, x_t] + b_h) \quad (23)$$

where $\odot$ denotes elementwise multiplication, $\tanh$ denotes the hyperbolic tangent function, and W is the trainable weight matrix.

z_t is the update gate, for the current hidden state h_t how many hidden states h_{t-1} and candidate states $\tilde{h}_t$ need to be retained are determined by z_t . Eq. (24) yields the update gate z_t .

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z) \quad (24)$$

where W_z is the trainable weight matrix.

The current hidden state h_t is obtained by a controlled combination of the last time step h_{t-1} and the current candidate state $\tilde{h}_t$ by updating the gate z_t . The formula for h_t is shown in (25).

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (25)$$

For the CNN channel, we use a one-dimensional convolutional neural network (CNN) is employed to capture local patterns and dependencies in the data. The CNN's core component is the convolutional layer, which extracts local features from the input data through convolutional kernels (filters). The most important feature of time series data is that it varies with the timeline, so we can think of it as one-dimensional data and there are local patterns on each time step which can be recognized and captured by a convolutional kernel. Thus, local features in time series can be captured and learned by CNN models and they can be utilized to

predict future trends. In addition, CNNs make the amount of model parameters small through the properties of parameter sharing and local connectivity, which not only reduces the possibility of model overfitting, but also improves the efficiency of computation. Time series are characterized by their sequentiality and temporal correlation, CNN extracts the correlation features between local time steps by 1D Convolution. The specific steps are as follows:

(1) Input Layer: The input data serves as a history window for the time series. Suppose we have an input of length T , where the dimension of the data at each time step is d ($d=1$ for univariate time series).

(2) Convolution Layer: A one-dimensional convolution kernel is applied on the time axis, and the sliding window extracts the same part of the features. The convolution operation is formulated as in (26):

$$y_i = \sum_{k=1}^K w_k \cdot x_{i+k-1} \quad (26)$$

Where, x_i , y_i , w_k denote the parameters of the convolution kernel for the input sequence, output sequence respectively, and K is used to denote the size of the convolution kernel.

Activation Layer: in this layer the ability of nonlinearity is introduced to the model, the activation function often used is RELU, the activation function formula for this layer is shown in (27):

$$f(x) = \max(0, x) \quad (27)$$

(3) Pooling Layer: Maximum pooling and average pooling are two commonly used pooling methods. This layer can be used for down sampling to reduce the data dimensions and avoid overfitting.

(4) Fully Connected Layer: Fully Connected Layer is used to map the data features that were pooled in the previous step in a specific way to get our predicted values.

(5) Output Layer: Predict the value of one or more future time steps.

$$output = activation(W \cdot input + b) \quad (28)$$

Eq. (28) shows the calculation process of the output layer. Where W is a weight matrix; $input$ denotes the input feature vector; b is the bias vector and $activation$ is the activation function.

Therefore, the two-channel hybrid network architecture allows each channel to focus on a different aspect of the data, meaning that the different channels will play to their respective strengths to capture different features and information in the data. The result of merging these channels results in a richer set of features, as well as the ability to learn more complex patterns and relationships in the data, which typically results in models with better performance and higher prediction accuracy.

4. Experiments

In this section, we detail six commonly used time series standard datasets and describe the evaluation criteria and benchmark models associated with the experiments. We demonstrate the effectiveness of the methodology proposed in this paper through extensive comparative experiments. Additionally, ablation studies were conducted to reveal the contribution of each component of the method to its overall performance. In addition, robustness analysis as well as experiments related to the visualization of the data were performed to demonstrate the performance of the D³CNet model.

4.1. Experimental data

In this paper, we use six of the most commonly used datasets in time series forecasting (e.g. Table 1) and conduct related experiments. All datasets used in this research experiment can be obtained from the resource library shared by the Autoformer on the Google Drive platform (https://drive.google.com/drive/folders/1ZOYpTUa82_jCcXldTmyr0LXQfvaM9vly). The description of each dataset used in this paper is as follows: the ETTh dataset, which contains four sub-datasets, namely two 15-minute-level datasets (ETTh1 and ETTh2) and two hour-level datasets (ETTh1 and ETTh2). Each of these datasets contains data on the load characteristics of seven different oil and power transformers for the time period from July 2016 to July 2018. The exchange_rate dataset, which covers daily foreign exchange rate movements for The exchange_rate dataset encompasses daily foreign exchange rate movements for eight countries from 1990 to 2016. The weather dataset comprises 21 meteorological indicators, including temperature and humidity, derived from observations recorded at 10-minute intervals in 2020. The relevant attributes of each dataset in the experiment are shown in Table 1.

4.2. Experimental settings

We tested the effectiveness of the D³CNet model on datasets from six widely used real-world application scenarios covering energy, economics, and weather prediction. The way each dataset is divided and the experiment-related parameter settings are shown in Table 2.

4.2.1. Evaluate Metrics

In order to evaluate the prediction performance of the model, we choose two commonly used metrics: Mean Square Error (MSE) and Mean Absolute Error (MAE), which is the average of the squared difference between the predicted values and the true values of the model. The smaller the MSE is, the smaller the error is, and the better the performance of the model will be. MAE is the absolute value of the difference between the predicted value and the true value, and then averaged; the smaller the MAE, the smaller the error, and the better the model performance. In order to evaluate the fit of the model to the data more comprehensively, we also use the coefficient of determination (R-squared, R²) to compare with the SparseTSF model. R2-Score is the variance between the predicted value and the true value of the model divided by the variance of the true value. The value of R2-Score can range from 0 to 1. A higher value of R² means a better fit to the data and a better prediction of the data. The higher the R² value, the better the fit of the model to the data and the stronger the predictive ability; on the contrary, the lower the R² value, the poorer the fit of the model and the weaker the predictive ability.

$$MSE = \frac{\sum_{i=1}^N (Predicted_i - Actual_i)^2}{N} \quad (29)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |Predicted_i - Actual_i| \quad (30)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (Predicted_i - Actual_i)^2 / N}{\sum_{i=1}^N (MeanOfActual - Actual_i)^2 / N} \quad (31)$$

Table 1
Basic information about the dataset.

Dataset	Dim	Input	Predict Length	Data size	Frequency
ETTh1	7	256	{96, 192, 336, 720}	17420	Hourly
ETTh2	7	256	{96, 192, 336, 720}	17420	Hourly
ETTm1	7	256	{96, 192, 336, 720}	69680	15 mins
ETTm2	7	256	{96, 192, 336, 720}	69680	15 mins
Weather	21	256	{96, 192, 336, 720}	52696	10 mins
Exchange	8	256	{96, 192, 336, 720}	7588	Daily

Table 2

Parameter settings for experiments with different data sets.

Dataset	Percentage of data	optimizer	Loss	learning_rate
ETTh1	6:2:2	Adam	MAE	0.0001
ETTh2	6:2:2	Adam	MAE	0.0001
ETTm1	6:2:2	Adam	MAE	0.0001
ETTm2	6:2:2	Adam	MAE	0.0001
Weather	7:1:2	Adam	MAE	0.001
Exchange	7:1:2	Adam	MAE	0.005

4.2.2. Baseline

We compared our proposed method with various advanced benchmark models in order to compare its advantages with other models. This comparison allows us to demonstrate the positive impact of the method on improving prediction accuracy. These models include SparseTSF (Lin et al., 2024), PatchTST (Nie et al., 2022), TimeMixer (Wang et al., 2024), DLinear (Zeng et al., 2023), Linear.

SparseTSF: The prediction task is mainly carried out by decoupling the trend and periodicity of time series data, using sparse prediction methods across time periods. Specifically, the technique focuses on cross-period trend forecasting by down sampling the original series, thus effectively extracting periodic features.

TimeMixer: By decomposing multiple scale mixing operations and obtain the complex patterns in data, the goal of improving prediction accuracy is achieved, which is a relatively novel time series prediction architecture.

PatchTST: It mainly regards time series as 64 labels composed of multiple variables, and improves the prediction performance of time series through self supervised learning and patch segmentation.

DLinear: Decomposes the data into trend and residuals and models them separately using a two-layer linear network, which is able to better obtain the trend and residual information of the time series. In addition, the problem of error accumulation in long time prediction is avoided by directly predicting multiple future time steps instead of using traditional autoregressive methods.

4.3. Comparison test

In Table 3, Table 4, Table 5, Table 6, Tables 7 and 8 display the results of multivariate long-term sequence prediction, with lower MSE and MAE indicating better predictive performance. The best results are displayed in bold black, and the second-best results are underlined. The data input length is I=720, and the predicted output length is O=(96,192,336,720).

In the multivariate time series forecasting task, Table 3-Table 6 show the results of comparing D³CNet with other models on the four ETT datasets. Table 3 demonstrates the comparison results of D³CNet with other models on the ETTh1 dataset. Table 3 shows that D³CNet has the lowest MSE and MAE at four different prediction steps, where the

Table 3

Multivariate time series prediction results for the ETTh1 dataset.

Models	Metric	Prediction Length				
		96	192	336	720	AVG
DLinear(2023)	MSE	0.072	0.094	0.118	0.228	0.128
	MAE	0.206	0.239	0.272	0.397	0.278
PatchTST(2023)	MSE	<u>0.059</u>	0.085	<u>0.077</u>	0.136	0.089
	MAE	<u>0.191</u>	0.234	<u>0.221</u>	0.293	0.235
Linear(2023)	MSE	0.089	0.096	0.109	0.210	0.126
	MAE	0.229	0.241	0.260	0.380	0.277
TimeMixer(2024)	MSE	0.196	<u>0.083</u>	0.089	0.114	0.088
	MAE	0.199	0.227	0.239	0.269	0.233
SparseTSF(2024)	MSE	0.062	0.196	0.087	<u>0.104</u>	<u>0.082</u>
	MAE	0.196	0.199	0.237	<u>0.253</u>	<u>0.226</u>
Ours	MSE	0.027	0.018	0.049	0.019	0.028
	MAE	0.118	0.098	0.177	0.104	0.124

Table 4

Multivariate time series prediction results for the ETTh2 dataset.

Models	Metric	Prediction Length				
		96	192	336	720	AVG
DLinear(2023)	MSE	0.146	0.177	0.194	0.270	0.197
	MAE	0.298	0.344	0.355	0.419	0.354
PatchTST(2023)	MSE	0.217	0.274	0.278	0.262	0.258
	MAE	0.368	0.414	0.419	0.408	0.402
Linear(2023)	MSE	0.145	0.178	0.195	0.248	0.191
	MAE	0.297	0.335	0.357	<u>0.401</u>	0.347
TimeMixer(2024)	MSE	<u>0.132</u>	<u>0.166</u>	<u>0.181</u>	0.251	<u>0.183</u>
	MAE	<u>0.284</u>	<u>0.324</u>	<u>0.344</u>	<u>0.401</u>	<u>0.338</u>
SparseTSF(2024)	MSE	0.148	0.175	0.187	<u>0.246</u>	0.189
	MAE	0.302	<u>0.335</u>	0.350	0.400	0.347
Ours	MSE	0.097	0.154	0.128	0.187	0.142
	MAE	0.263	0.363	0.313	<u>0.401</u>	0.335

Table 5

Multivariate time series prediction results for the ETTm1 dataset.

Models	Metric	Prediction Length				
		96	192	336	720	AVG
DLinear(2023)	MSE	0.026	0.05	0.06	0.086	0.056
	MAE	0.122	0.169	0.184	0.22	0.174
PatchTST(2023)	MSE	0.03	0.045	0.055	0.072	0.051
	MAE	0.133	0.161	0.178	0.208	0.17
Linear(2023)	MSE	0.027	<u>0.04</u>	0.06	0.081	0.052
	MAE	0.124	0.15	0.187	0.215	0.169
TimeMixer(2024)	MSE	<u>0.027</u>	0.041	0.053	0.07	<u>0.048</u>
	MAE	0.125	0.153	0.175	0.202	0.164
SparseTSF(2024)	MSE	0.026	0.039	<u>0.052</u>	<u>0.07</u>	0.047
	MAE	0.123	<u>0.151</u>	<u>0.174</u>	<u>0.201</u>	<u>0.162</u>
Ours	MSE	0.039	0.082	0.033	0.069	0.056
	MAE	0.108	0.156	0.113	0.17	0.137

Table 6

Multivariate time series prediction results for the ETTm2 dataset.

Models	Metric	Prediction Length				
		96	192	336	720	AVG
DLinear (2023)	MSE	<u>0.063</u>	0.100	0.127	0.183	0.118
	MAE	<u>0.183</u>	0.241	0.274	0.333	0.258
PatchTST (2023)	MSE	0.071	0.109	<u>0.116</u>	0.181	0.119
	MAE	0.198	0.253	0.265	0.331	0.262
Linear(2023)	MSE	0.067	0.095	0.132	0.186	0.12
	MAE	0.191	0.234	0.28	0.334	0.26
TimeMixer (2024)	MSE	0.069	0.099	0.124	<u>0.171</u>	0.116
	MAE	0.196	0.241	0.271	0.326	0.259
SparseTSF (2024)	MSE	0.065	<u>0.095</u>	0.122	0.172	<u>0.114</u>
	MAE	0.187	<u>0.233</u>	<u>0.267</u>	<u>0.323</u>	<u>0.253</u>
Ours	MSE	0.033	0.055	0.098	0.132	0.08
	MAE	0.16	0.222	0.277	0.311	0.243

average MSE is improved by 45 % and the average MAE is improved by 65 %, where the average MSE is improved by 45 % and the average MAE is improved by 65 %. From, Tables 4, 5, and 6 we see that D³CNet still achieves better results for MAE and MSE in the remaining three ETT datasets.

4.3.1. Multivariate results

Table 3-Table 8 show the results of the comparison between D³CNet and other models. In the multivariate time series prediction task, our method demonstrates superior performance on six benchmark datasets, as evidenced by our experimental results. The experimental results demonstrate the large improvement of D³CNet's results on six datasets when performing multivariate prediction, and our method improves the average MSE by more than 20 % and reduces the MAE by 15.1 % on the four ETT datasets compared to SparseTSF. On the Weather dataset, our

Table 7

Multivariate time series prediction results for the Weather dataset.

Models	Metric	Prediction Length				
		96	192	336	720	AVG
DLinear(2023)	MSE	0.0056	0.0061	0.0064	0.0070	0.0060
	MAE	0.0614	0.0655	0.0655	0.0655	0.0640
PatchTST(2023)	MSE	0.0013	0.0027	0.0031	0.0022	0.0023
	MAE	0.0267	0.0384	0.0435	0.0349	0.0359
Linear(2023)	MSE	0.0060	0.0064	0.0070	0.0064	0.0060
	MAE	0.6530	0.0675	0.0721	0.0673	0.2150
TimeMixer(2024)	MSE	0.0013	0.0015	<u>0.0017</u>	0.0022	0.0020
	MAE	0.0274	0.2958	0.0316	0.0361	0.0980
SparseTSF(2024)	MSE	<u>0.0012</u>	<u>0.0014</u>	0.0016	<u>0.0021</u>	<u>0.0020</u>
	MAE	<u>0.0258</u>	<u>0.0282</u>	0.0301	<u>0.0349</u>	<u>0.0300</u>
Ours	MSE	0.0007	0.0012	<u>0.0017</u>	0.0017	0.0013
	MAE	0.0209	0.0254	<u>0.0305</u>	0.0301	0.0267

Table 8

Multivariate time series prediction results for Exchange_rate dataset.

Models	Metric	Prediction Length				
		96	192	336	720	AVG
DLinear(2023)	MSE	0.109	0.272	0.496	2.504	0.845
	MAE	0.254	0.406	0.527	1.394	0.645
PatchTST(2023)	MSE	<u>0.104</u>	<u>0.198</u>	<u>0.384</u>	1.019	<u>0.426</u>
	MAE	<u>0.242</u>	<u>0.344</u>	<u>0.479</u>	0.773	<u>0.46</u>
Linear(2023)	MSE	0.276	0.397	0.715	2.293	0.92
	MAE	0.448	0.524	0.665	1.268	0.726
TimeMixer(2024)	MSE	0.160	0.302	0.533	<u>0.819</u>	0.454
	MAE	0.297	0.419	0.578	<u>0.719</u>	0.503
SparseTSF(2024)	MSE	0.153	0.271	0.429	1.067	0.48
	MAE	0.301	0.411	0.516	0.828	0.514
Ours	MSE	0.062	0.140	0.219	0.320	0.185
	MAE	0.215	0.317	0.393	0.518	0.360

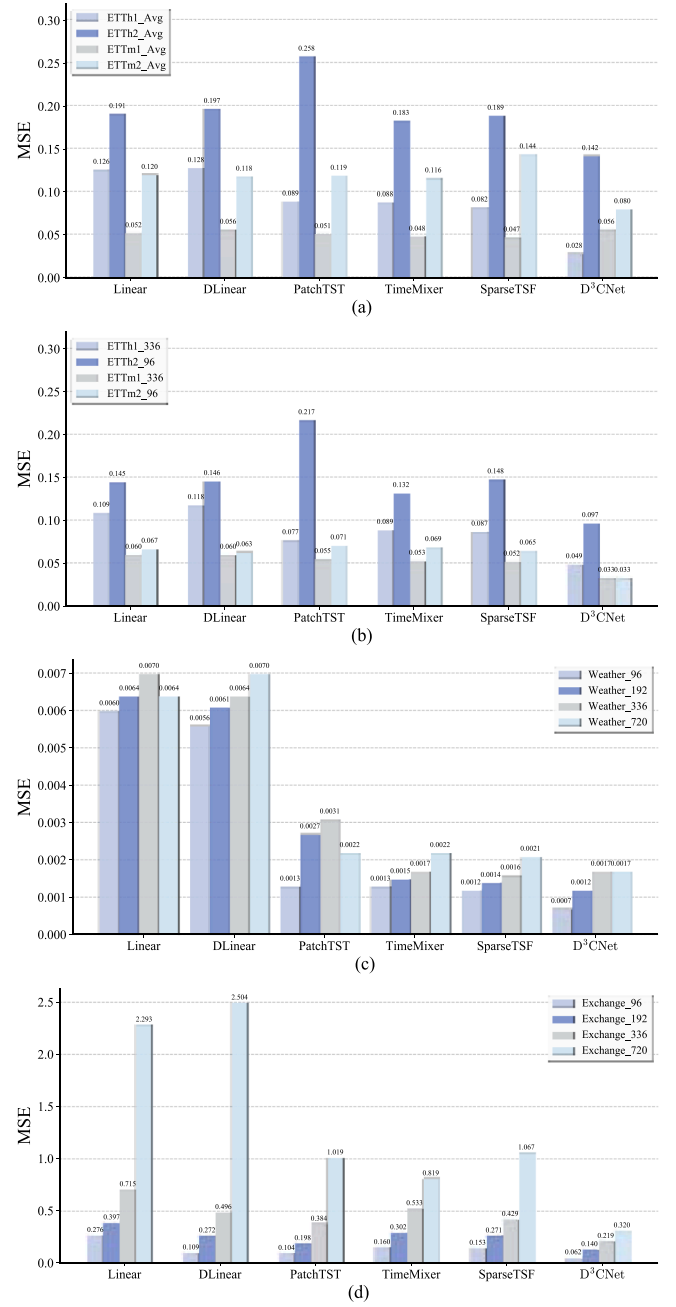
method shows significant improvement, with a 35 % reduction in the average MSE and a 11 % reduction in the average MAE. There is no significant seasonality in the Exchange_rate dataset, where our D³CNet is optimal at different prediction steps compared to the Transformer-based PatchTST. The average MSE and average MAE are reduced by 56.57 % and 21.74 %, respectively. These results show that we are able to handle long-term time series forecasting tasks in various domains in practical applications.

In order to better see the prediction effect of D³CNet, as shown in Fig. 8.(d), on the Exchange_rate dataset, our D³CNet achieves the optimal effect at different prediction steps. We can see from the Fig. that when the prediction step size is 720, the predictive performance of the remaining models decrease significantly, while D³CNet still manages to maintain a low MSE, which further illustrates the superior predictive performance of D³CNet in predicting long time series.

Table 9 presents the mean and standard deviation results of model MAE for six datasets. It can be observed that, except for the ETTm1 dataset, the MAE standard deviation to mean ratios of the other five datasets are all below 0.3, which further confirms the good predictive stability of the model. For the ETTm1 dataset, the ratio of MAE standard deviation to mean is in the range of 0.27-0.36. This fluctuation may be due to the noise interference and data distribution characteristics of the dataset itself, which may affect the model prediction to some extent. But overall, this proportion is controlled within 0.4 for all datasets, which is within a controllable and reasonable range.

4.3.2. Univariate results

Four tables, Tables 10, 11, 12, and 13 show how D³CNet compares with other best methods in terms of univariate prediction on the ETT dataset. In most cases, D³CNet outperforms the other methods with an average MSE improvement of 42.2 % and an average MAE improvement of 23.3 % compared to the best results in the tables. The improvement is

**Fig. 8.** Comparison of assessment indicators for the six datasets.**Table 9**

Mean-Std Dev MAE experimental table.

Dataset	Metric	Prediction Length			
		96	192	336	720
ETTh1	Mean	0.131	0.102	0.180	0.122
	±Std	±0.036	±0.29	±0.047	±0.028
ETTh2	Mean	0.088	0.156	0.118	0.159
	±Std	±0.025	±0.045	±0.044	±0.059
ETTh1	Mean	0.277	0.383	0.375	0.362
	±Std	±0.049	±0.040	±0.050	±0.035
ETTh2	Mean	0.186	0.271	0.247	0.318
	±Std	±0.044	±0.039	±0.025	±0.039
Weather	Mean	0.028	0.021	0.035	0.0323
	±Std	±0.006	±0.006	±0.003	±0.040
Exchange	Mean	0.250	0.380	0.447	0.527
	±Std	±0.024	±0.038	±0.066	±0.049

Table 10

Univariate time series prediction results for the ETTh1 dataset.

Models	Metric	Prediction Length				
		96	192	336	720	AVG
DLinear(2023)	MSE	0.099	0.126	0.148	0.236	0.152
	MAE	0.246	0.277	0.306	0.395	0.306
Linear(2023)	MSE	0.119	0.119	0.159	0.223	0.155
	MAE	0.274	0.272	0.318	0.387	0.313
SparseTSF(2024)	MSE	<u>0.064</u>	<u>0.077</u>	<u>0.088</u>	<u>0.105</u>	<u>0.084</u>
	MAE	<u>0.198</u>	<u>0.219</u>	<u>0.238</u>	<u>0.254</u>	<u>0.227</u>
Ours	MSE	0.033	0.043	0.046	0.037	0.04
	MAE	0.148	0.173	0.177	0.147	0.161

Table 11

Univariate time series prediction results for the ETTh2 dataset.

Models	Metric	Prediction Length				
		96	192	336	720	AVG
DLinear(2023)	MSE	0.146	0.177	0.194	0.27	0.197
	MAE	0.298	0.334	0.356	0.42	0.352
Linear(2023)	MSE	0.145	0.177	0.195	0.248	0.191
	MAE	<u>0.297</u>	0.335	0.357	0.401	0.348
SparseTSF(2024)	MSE	0.149	<u>0.176</u>	<u>0.188</u>	<u>0.246</u>	<u>0.190</u>
	MAE	0.303	<u>0.336</u>	<u>0.351</u>	<u>0.401</u>	<u>0.348</u>
Ours	MSE	0.073	0.088	0.068	0.110	0.085
	MAE	0.248	0.263	0.231	0.305	0.262

Table 12

Univariate time series prediction results for the ETTm1 dataset.

Models	Metric	Prediction Length				
		96	192	336	720	AVG
DLinear(2023)	MSE	0.035	0.05	<u>0.06</u>	0.086	0.058
	MAE	0.144	0.169	0.184	0.22	0.179
Linear(2023)	MSE	0.035	0.047	<u>0.06</u>	0.081	0.056
	MAE	0.143	0.166	0.187	0.215	0.178
SparseTSF(2024)	MSE	0.027	0.039	0.052	<u>0.071</u>	0.047
	MAE	0.125	0.151	0.174	0.204	0.164
Ours	MSE	<u>0.034</u>	<u>0.044</u>	0.062	0.066	<u>0.052</u>
	MAE	0.125	0.146	0.164	0.177	0.153

Table 13

Univariate time series prediction results for the ETTm2 dataset.

Models	Metric	Prediction Length				
		96	192	336	720	AVG
DLinear(2023)	MSE	0.072	0.100	0.127	0.183	0.121
	MAE	0.202	0.241	0.274	0.333	0.263
Linear(2023)	MSE	0.072	0.105	0.132	0.186	0.124
	MAE	0.200	0.246	0.280	0.334	0.265
SparseTSF(2024)	MSE	<u>0.067</u>	<u>0.097</u>	<u>0.125</u>	<u>0.175</u>	<u>0.116</u>
	MAE	<u>0.191</u>	<u>0.237</u>	<u>0.272</u>	<u>0.327</u>	<u>0.257</u>
Ours	MSE	0.056	0.086	0.097	0.057	0.074
	MAE	0.158	0.223	0.218	0.160	0.190

especially significant on ETTh compared to SparseTSF, with an average MSE improvement of 54.7 % and an average MAE improvement of 26.4 %. An average improvement of 24.8 % is also realized on the ETTm dataset. Overall, D³CNet also shows a more significant performance improvement in univariate prediction compared to the best methods above, thus also demonstrating the better generalization of D³CNet.

4.4. Ablation experiments

Three ablation experiments were conducted on the ETTh2, ETTm2, and Exchange_rate datasets to evaluate the role of the S-EMD module and the dual-channel hybrid network in the D³CNet model. Three

comparison experiments were performed on the original D³CNet model. In the w/o S-EMD experiment, the secondary decomposition part of the model was removed. In the w/o conv1D experiment, we replaced the dual-channel Convolution module with the GRU model. Finally, in the w/o GRU experiments, we replaced the dual-channel GRU model with the Convolution module.

The following conclusions can be drawn from Table 14:

- (1) After removing the S-EMD module, the prediction of the model decreases dramatically, which suggests that this module can facilitate multi-scale feature extraction in our model and thus improve the accuracy of prediction.
- (2) Removing conv1D has the most significant effect on the ETTh2 and ETTm2 datasets, indicating that, for data exhibiting periodicity, our conv1D is adept at capturing dependencies between temporal features with greater efficacy.
- (3) The GRU channel is mainly responsible for obtain long-term dependencies in the timing data. After removing the GRU, the long-term dependencies cannot be effectively captured for the Exchange_rate dataset without periodicity, which leads to a large bias in the prediction results. Conclusion (2)(3) not only proves the effectiveness of the dual-channel hybrid network structure, but also shows that choosing the appropriate model for different data patterns for prediction helps to improve the model performance.

Therefore, summarizing the above conclusions, it can be seen that the inter-combination of components in the D³CNet model makes our prediction results more accurate. The multiscale features are extracted by the S-EMD module, and then the deep temporal features are extracted by the conv1D in the dual-channel hybrid network, while the GRU captures the potential long-term dependence information of the temporal data, and finally the predicted sequence closer to the real value is obtained by the feature fusion of the two channels. The above ablation experiments once again demonstrate the influence and role of each component in D³CNet, and further confirm the superiority of the model in theory and practice.

D³CNet shows the lowest MSE and MAE on all three different datasets. Figs. 9–11, provide a more intuitive view of the impact of each component on D³CNet. We find that removing any of the components of D³CNet significantly decreases the prediction of the model, and different components have different impacts on different datasets. As shown in Fig. 10.(a) and Fig. 11.(a), for the ETTm2 dataset, the removal of conv1D has the most significant impact on the model performance, while for the Exchange dataset, it is the removal of the component GRU that has the greatest effect on the predictive performance of the model. This illustrates that different components affect the D³CNet model to different degrees, and further illustrates the rationality and effectiveness of the D³CNet model.

4.5. Experimental analysis

We performed multivariate time series prediction experiments using multiple models on the ETTh2 dataset and visualized a comparison of the last set of data predicted with a prediction step size of 336 steps. This allows us to clearly compare the different models. As can be seen in Fig. 12, each model gives better access to the whole trend of the data. However, for certain local variation features in the data, the performance of the other models is relatively weak. This indicates that these models are more inclined to focus on the global features and trends of the time series data, and tend to ignore the local features of the data, which affects the fitting effect of the models. In addition, when the prediction range is too long, the prediction effect of the other models also decreases significantly. Therefore, D³CNet predicts best and most accurately on datasets with significant distributional bias (ETTh2). And the fitting effect in Fig. 12.(a) further proves that D³CNet can mine deeper information in the data and model it.

Table 14
Results of ablation experiments.

Datasets		ETTh2				ETTm2				Exchange_rate			
Prediction length		96	192	336	720	96	192	336	720	96	192	336	720
w/o S-EMD	MSE	0.154	0.226	0.253	0.286	0.335	0.407	0.369	0.354	0.208	0.172	0.370	0.501
	MAE	0.337	0.420	0.449	0.483	0.457	0.487	0.464	0.477	0.361	0.384	0.533	0.596
w/o conv1D	MSE	0.199	0.272	0.146	0.227	1.095	0.889	0.749	0.729	0.265	0.335	0.271	0.643
	MAE	0.398	0.280	0.328	0.437	1.017	0.908	0.831	0.797	0.453	0.490	0.451	0.608
w/o GRU	MSE	0.160	0.243	0.148	0.186	0.466	0.395	0.378	0.358	0.231	0.336	0.484	0.516
	MAE	0.362	0.456	0.354	0.404	0.549	0.501	0.494	0.459	0.443	0.550	0.656	0.702
D³CNet (Ours)	MSE	0.097	0.154	0.128	0.187	0.033	0.055	0.098	0.132	0.062	0.140	0.219	0.320
	MAE	0.263	0.363	0.313	0.401	0.160	0.222	0.277	0.311	0.215	0.317	0.393	0.518

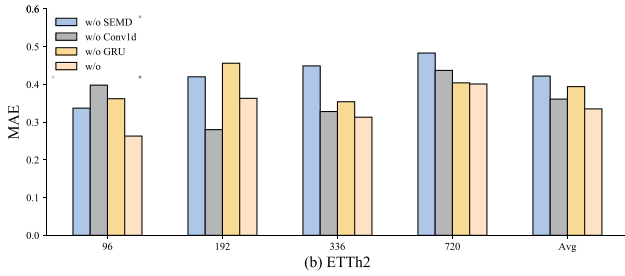
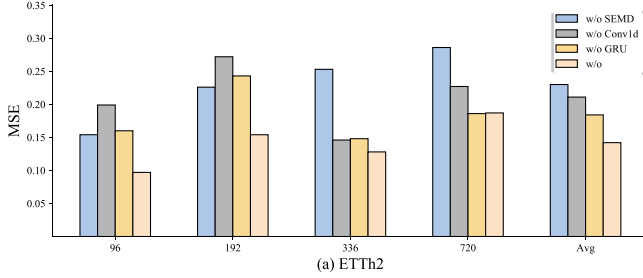


Fig. 9. Experimental results for the ETTh2 dataset.

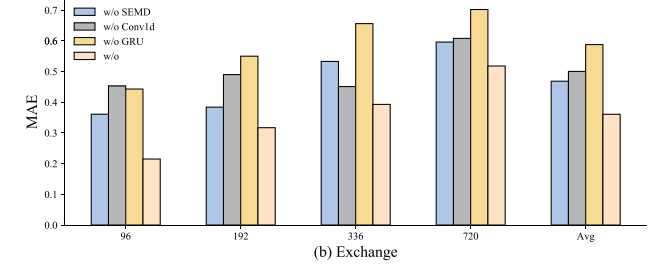
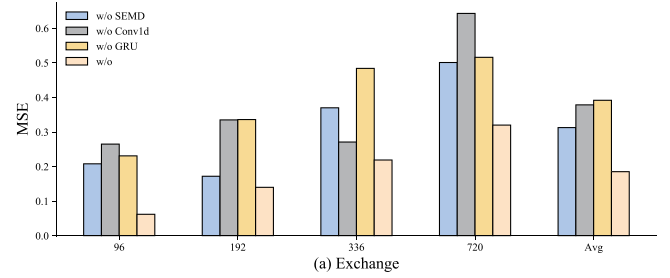


Fig. 11. Experimental results for Exchange dataset.

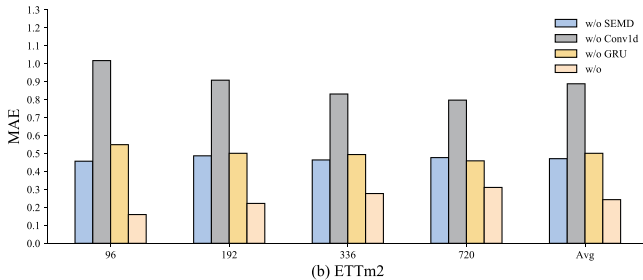
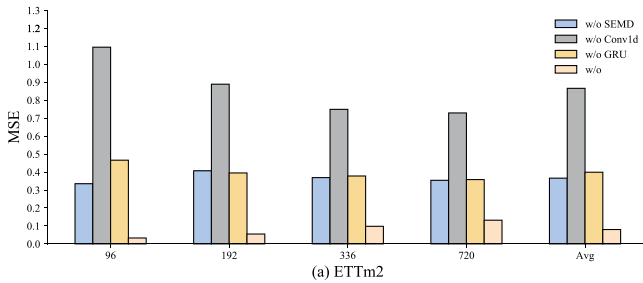


Fig. 10. Experimental results for the ETTm2 dataset.

In the multivariate time series prediction task, comparing the results of Tables 3 and 5, it can be found that when the prediction step size is 192 (about 1 week) and 720 (about 1 month), the MAE and MSE

indicators of the ETTm1 dataset are higher than those of the ETTh1 dataset. We speculate that the root cause of this prediction bias lies in the difference in frequency characteristics and period matching between the two datasets: on the one hand, the proportion of interference fluctuations in the data at different sampling frequencies is different (such as ETTm1's high-frequency data containing more instantaneous noise); On the other hand, different prediction steps correspond to different time periods, and the adaptability of the model to "frequency period" will directly affect the prediction performance.

To verify this hypothesis, we conducted comparative experiments with different sampling frequencies based on ETTm1 to explore the regulatory effect of prediction step size and frequency on interference fluctuations.

The ettm-30 and ettm-45 datasets in Table XX are both generated by sampling from the original ETTm1 data, corresponding to sampling frequencies of 30 minutes and 45 minutes, respectively. To facilitate a more intuitive analysis of the impact of frequency on prediction performance, ettm-15 directly utilizes the ETTm1 dataset with a 15-minute frequency, while ettm-60 directly employs the ETTm1 dataset with a 60-minute frequency for experimental analysis.

Table 15, the results showed that at 192 and 720 step sizes, the MAE index showed a continuous optimization trend as the sampling frequency decreased (granularity became coarser) - for example, in the 192 step size, the MAE of ettm-60 was significantly lower than that of ettm-15; In 720 step size, the MAE of ettm-60 (0.104) is also better than that of ettm-15.

This result confirms the core conclusion: the model has stronger adaptability to "low-frequency data - long cycle step size" - ETTh1 itself

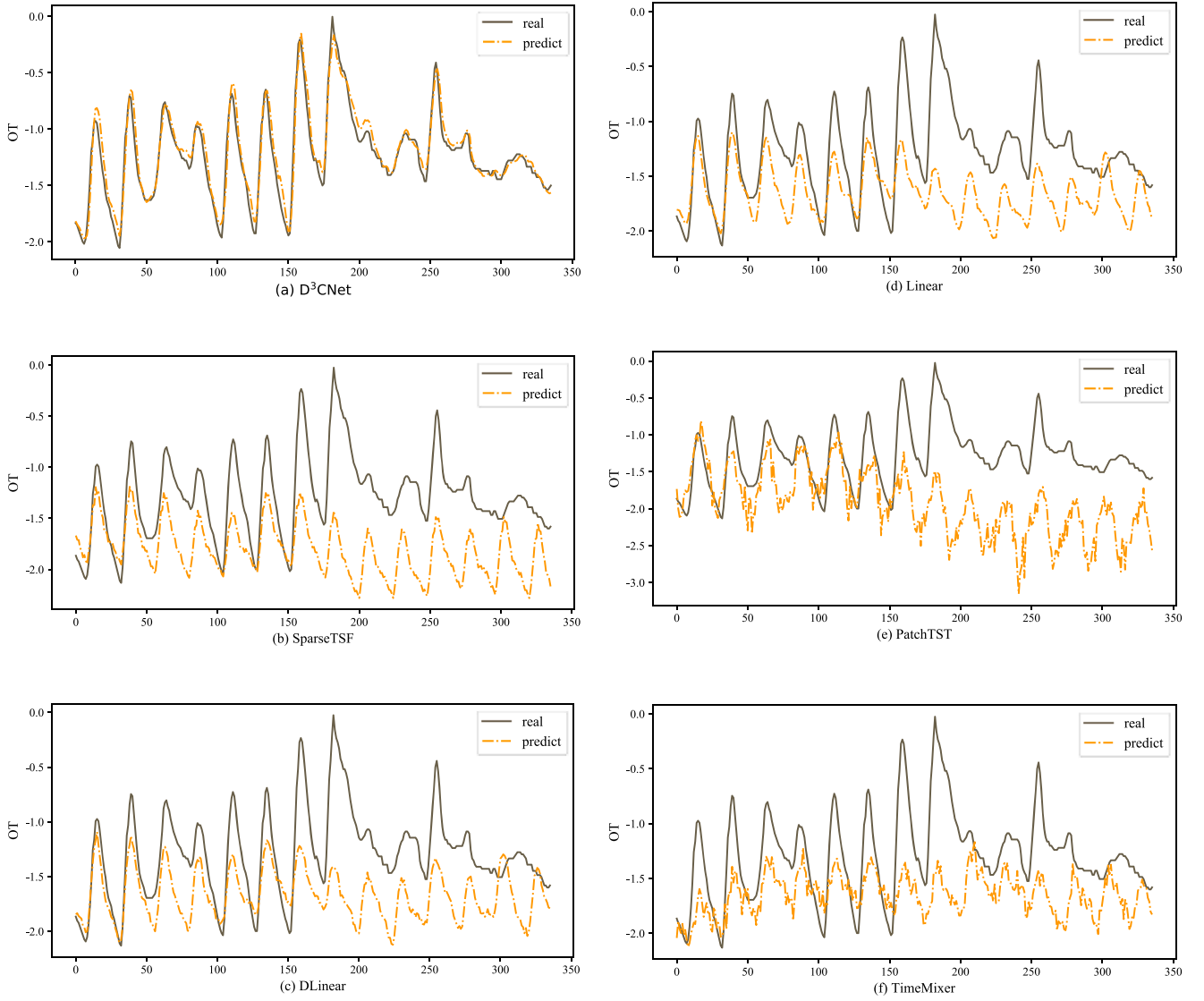


Fig. 12. Plot of true vs. predicted values for the ETTh2 dataset.

Table 15
Sampling Analysis of ETTm1.

Dataset	Metric	Prediction Length			
		96	192	336	720
ettm-60	MAE	0.118	0.098	0.177	0.104
ettm-45	MAE	0.098	0.119	0.124	0.108
ettm-30	MAE	0.075	0.126	0.112	0.120
ettm-15	MAE	0.108	0.156	0.113	0.170

is a 1-hour sampling (low-frequency), which is more compatible with time units of 192 (weekly cycle) and 720 (monthly cycle), with less interference fluctuations and clearer cycle characteristics; ETTm1 is a high-frequency sampling method, and its interference fluctuations can blur the long-period pattern, ultimately leading to lower prediction metrics than ETTh1.

4.5.1. Visual analysis

We applied D³CNet and SparseTSF on the weather dataset to compare the multivariate time series prediction results with different prediction lengths. As shown in Table 16, the improvement in R² is significant despite the fact that the two evaluated metrics, MAE and

Table 16
Comparison of evaluation indicators between D³CNet and SparseTSF in the Weather dataset.

Models	Metric	Prediction Length				
		96	192	336	720	AVG
SparseTSF (2024)	MSE	0.0012	0.0014	0.0016	0.0021	0.0020
	MAE	0.0258	0.0282	0.0301	0.0349	0.0300
	R ²	0.33	0.21	0.11	0.15	0.2
Ours	MSE	0.0007	0.0012	0.0017	0.00007	0.0010
	MAE	0.0209	0.0254	0.0305	0.0044	0.0200
	R ²	0.71	0.52	0.34	0.76	0.58

MSE, of the model on this dataset are very different. This indicates that the D³CNet model performs better in fitting the data and is able to better capture the potentially important information in the data. The comparison of the true and predicted values of the D³CNet and SparseTSF models for different prediction steps on the Weather dataset are shown in Figs. 13 and 14, which show that the D³CNet model is better at predicting the upward and downward turning points, and both achieve lower error and higher data fit, further proving the modeling and prediction ability of D³CNet.

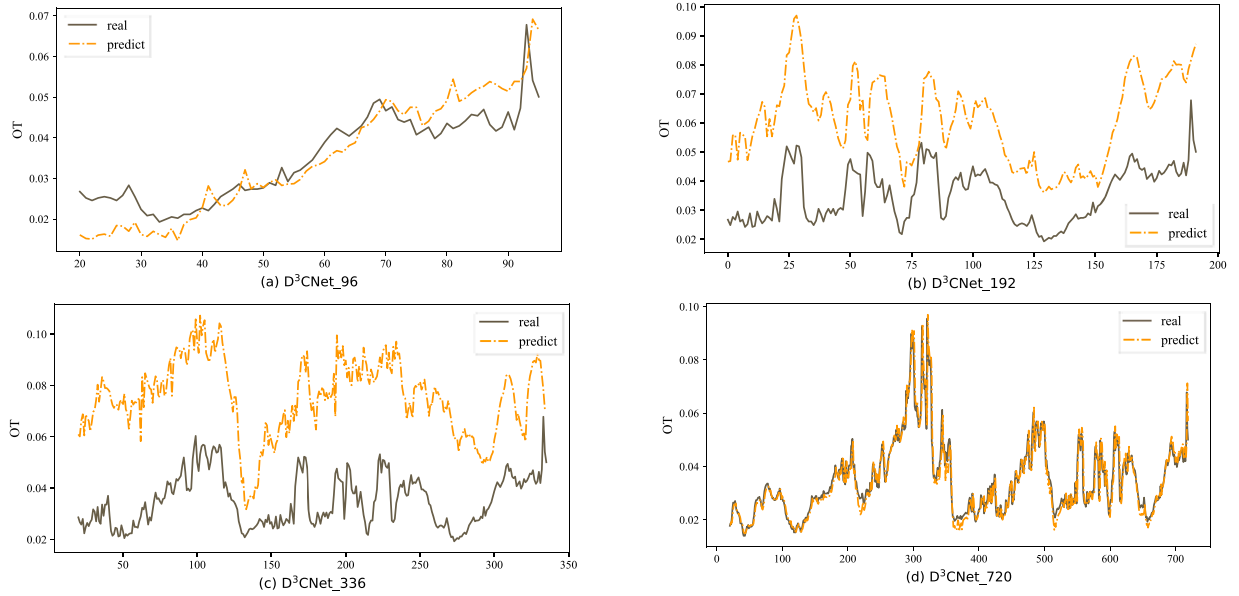
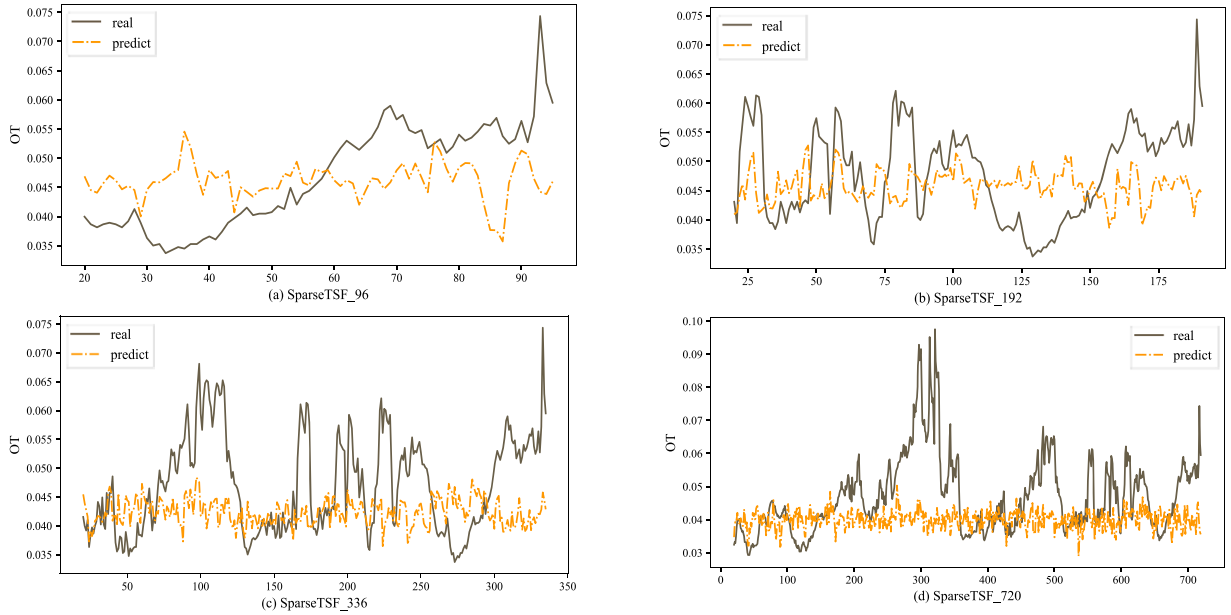
Fig. 13. Plot of true vs. predicted values of D³Cnet.

Fig. 14. Plot of true vs. predicted values of SparseTSF.

4.5.2. Robustness analysis

In this paper, we demonstrate the robustness of the model by injecting Gaussian noise, that is, $0.1 \cdot N(0,1)$ Gaussian noise is injected into the original sequences and four sets of experiments are performed in the training phase with and without noise, respectively. These

experiments were performed on ETTh1. YES and NO are used in the table to indicate the presence or absence of noise, respectively.

The results are shown in Table 17. We introduced Gaussian noise in each of the above four cases, and the impact on the D³Cnet performance was relatively small, as evidenced by the fact that the average MSE

Table 17

Experimental results for the ETTh1 dataset with and without noise.

Training(Noise)	NO				YES			
	NO		YES		NO		YES	
Testing(Noise)								
Metrics	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	0.026	0.110	0.031	0.115	0.027	0.118	0.020	0.103
	0.018	0.097	0.023	0.104	0.018	0.099	0.027	0.125
	0.045	0.178	0.049	0.169	0.049	0.178	0.049	0.181
	0.025	0.106	0.019	0.105	0.020	0.105	0.020	0.103

decreased by less than 3 %, while the impact on the average MAE was even smaller, decreasing by less than 0.5 %. Thus, when using the D³CNet model for prediction, the impact on both the MAE and MSE of the prediction results is low when a certain amount of noise is mixed into the data input, and Fig. 15.(a) better visualizes that the added noise has very little impact on our MAE, which suggests that the model exhibits better robustness.

4. Conclusion

In this paper, we propose a long time series forecasting method based on quadratic decomposition and dual-channel hybrid network, named D³CNet, which integrates quadratic decomposition technique and dual-channel hybrid network. The model greatly improves the extraction of multi-scale features by using several different decompositions means to decompose the time series into multiple stable modal components with different dominant frequencies and constructing seasonal component components. We then also introduced a dual-channel hybrid network for modeling, allowing each channel to focus on different features of the data. The first channel reveals the implicit dependencies in the time series more effectively through data normalization and GRU (Gated Recurrent Unit). The second channel, on the other hand, utilizes CNN (Convolutional Neural Network) models to capture local and seasonal features of the time series. Ultimately, the prediction results of the two channels are integrated to enable the modeling of the time series. We also conducted a systematic analysis of D³CNet's computational complexity. Taking ETTh1 as an example, the average number of model parameters is 44,581, with a training time of approximately 150 seconds and a prediction time of approximately 70 milliseconds.

A large number of experimental results show that our proposed prediction method based on quadratic decomposition and dual-channel hybrid network exhibits high accuracy with smaller prediction errors in

a variety of long-term time series prediction tasks. This is more conducive to relevant decision making. This proves the effectiveness and practicality of the D³CNet long time series forecasting method proposed in this paper.

Although this paper has made significant progress in improving the model's ability to extract multi-scale features, there are still deficiencies in multi-dimensional feature screening. The model fails to effectively eliminate useless features when extracting features, resulting in increased complexity and reduced training efficiency. To address this problem, future work will do further research in terms of feature screening. The current feature screening methods have limitations in dealing with high-dimensional data, and in the future, feature selection algorithms based on information entropy, mutual information, or sparse representation can be introduced to retain only those features that contribute significantly to the model performance by quantifying the importance of the features. Second, the introduction of regularization techniques will help reduce the interference of useless features. Techniques such as L1 regularization can achieve feature sparsification by automatically suppressing the weights of unimportant features, thus reducing model complexity and improving generalization. Finally, feature sharing mechanisms under multi-task learning frameworks will be explored to achieve feature co-optimization between different tasks. By sharing features, the model can learn simultaneously in multiple tasks, thus improving feature utilization and model generalization ability.

In summary, future research will focus on the optimization of feature screening mechanisms, the application of regularization techniques, and the exploration of multi-task learning. These research directions are not only expected to solve the current problems of high model complexity and low training efficiency, but also provide new ideas and methods for the further development of multi-scale feature extraction models.

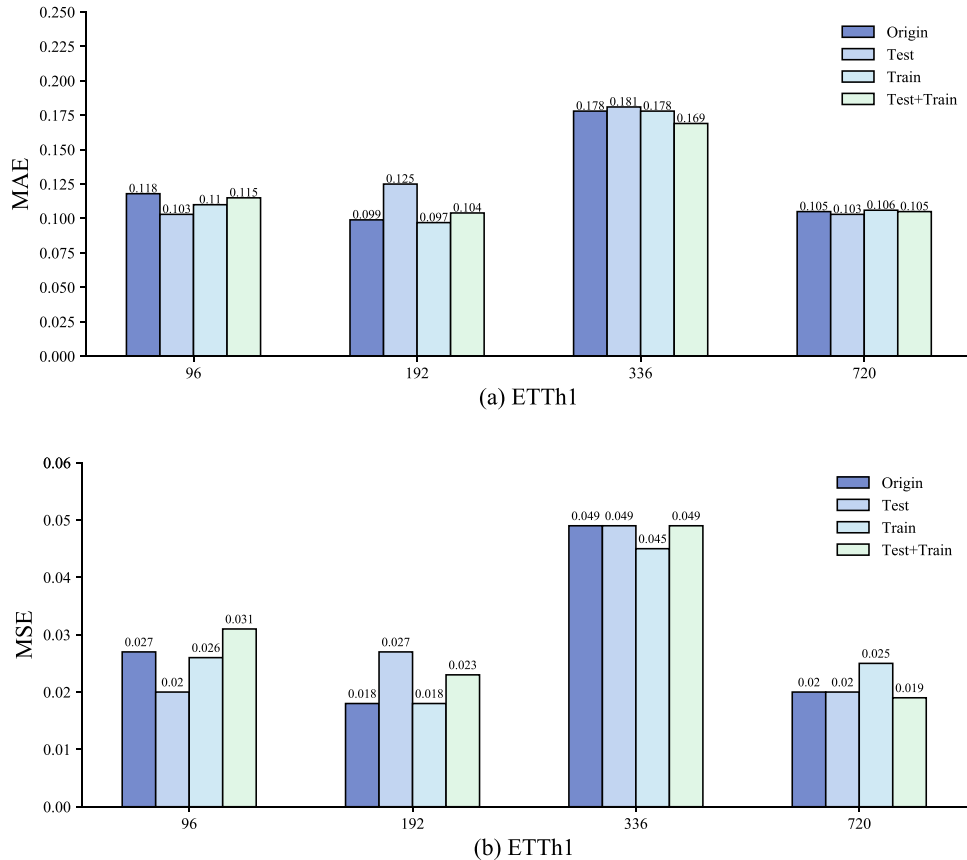


Fig. 15. Experimental results for the ETTh1 dataset with and without noise.

CRediT authorship contribution statement

Bo He: Supervision. **Yunya Bo:** Writing – review & editing, Writing – original draft, Resources, Project administration, Data curation. **Longbing Li:** Visualization.

Declaration of competing interest

We hereby declare that the manuscript entitled “D3CNet: Long-Term Time Series Prediction Based on Dual Decomposition and Dual-Channel Hybrid Network” is original, has not been previously published, and is not currently under consideration for publication elsewhere.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

We understand that the Corresponding Author is the sole contact for the Editorial process. He is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs.

Acknowledgements

This work is funded by the humanities and social sciences research project of the ministry of education (No.24YJA870003) and the Chongqing University of Technology Graduate Education High Quality Development Project (No.gzlcx20253209)

At this point, we would like to express our gratitude to the journal editors, reviewers, and all relevant personnel who participated in the peer review of this article.

References

- Chaovalit, P., Gangopadhyay, A., Karabatis, G., et al. (2011). Discrete wavelet transform-based time series analysis and mining. *ACM Computing Surveys (CSUR)*, 43(2), 1–37.
- Faghih, S., Shah, A., Wang, Z., et al. (2020). Taxi and mobility: modeling taxi demand using ARMA and linear regression. *Procedia Computer Science*, 177, 186–195.
- Feng, Z., & Niu, W. (2021). Hybrid artificial neural network and cooperation search algorithm for nonlinear river flow time series forecasting in humid and semi-humid regions. *Knowledge-Based Systems*, 211, Article 106580.
- Gong, H., & Xing, H. (2024). Predicting the highest and lowest stock price indices: A combined BiLSTM-SAM-TCN deep learning model based on re-decomposition. *Applied Soft Computing*, 167, Article 112393.
- He, B., Li, L., & Zhang, Q. (2025). PNC-based trend feature extraction method for time series data[J]. *Neurocomputing*, 618, Article 129174.
- Ilbert R, Odonnat A, Feofanov V, et al. Samformer: unlocking the potential of transformers in time series forecasting with sharpness-aware minimization and channel-wise attention. arXiv preprint arXiv:2402.10198, 2024.
- Jin, X. I. A., Zhengqun, WANG, & Shiming, Z. H. U. (2023). A traffic flow prediction model based on time series decomposition. *Computer Applications*, 43(4), 1129.
- Kong, W., Ju, Y., Zhang, S., et al. (2025). Graph enhanced spatial-temporal transformer for traffic flow forecasting[J]. *Applied Soft Computing*, 170, Article 112698.
- Lai, G., Chang, W C, Yang, Y., et al. (2018). Modeling long-and short-term temporal patterns with deep neural networks[C]. In *The 41st international ACM SIGIR conference on research & development in information retrieval* (pp. 95–104).
- Lin S, Lin W, Wu W, et al. Sparsetsf: modeling long-term time series forecasting with 1k parameters. arXiv preprint arXiv:2405.00946, 2024.
- Liu, M., Zeng, A., Chen, M., et al. (2022). Scinet: time series modeling and forecasting with sample convolution and interaction. *Advances in Neural Information Processing Systems*, 35, 5816–5828.
- Liu, T., Ma, X., Li, S., et al. (2022). A stock price prediction method based on meta-learning and variational mode decomposition. *Knowledge-Based Systems*, 252, Article 109324.
- Liu, S., Yu, H., Liao, C., et al. (2022). Pyraformer: low-complexity pyramidal attention for long-range time series modeling and forecasting[C]. In # *PLACEHOLDER_PARENT_METADATA_VALUE#*.
- Liu Y, Hu T, Zhang H, et al. itransformer: inverted transformers are effective for time series forecasting. arXiv preprint arXiv:2310.06625, 2023.
- Liu, J., Huang, X., Li, Q., et al. (2023). Hourly stepwise forecasting for solar irradiance using integrated hybrid models CNN-LSTM-MLP combined with error correction and VMD. *Energy Conversion and Management*, 280, Article 116804.
- Ma, X., Zhao, T., Guo, Q., et al. (2022). Fuzzy hypergraph network for recommending top-K profitable stocks. *Information Sciences*, 613, 239–255.
- Ma, X., Li, X., Fang, L., et al. (2024). U-Mixer: an unet-mixer architecture with stationarity correction for time series forecasting[C]. In , 38. *Proceedings of the AAAI Conference on Artificial Intelligence* (pp. 14255–14262).
- Matsubara, Y., Sakurai, Y., Van Panhuis W, G., et al. (2014). FUNNEL: automatic mining of spatially coevolving epidemics[C]. In *Proceedings of the 20th ACM SIGKDD international conference on Knowledge discovery and data mining* (pp. 105–114).
- Nie Y, Nguyen N H, Sinthong P, et al. A time series is worth 64 words: long-term forecasting with transformers. arXiv preprint arXiv:2211.14730, 2022.
- Oreshkin B N, Carpo D, Chapados N, et al. N-BEATS: neural basis expansion analysis for interpretable time series forecasting. arXiv preprint arXiv:1905.10437, 2019.
- Qin, H., Zhan, X., Li, Y., et al. (2021). Network-wide traffic states imputation using self-interested coalitional learning[C]. In *Proceedings of the 27th ACM SIGKDD conference on knowledge discovery & data mining* (pp. 1370–1378).
- Raghuvanshi, S. (2024). Assessing the impact of seasonal decomposition on the time series analysis accuracy: A comprehensive study. *Int J Res Appl Sci Eng Technol*, 10.
- Rhif, M., Abbas, A B, Martínez, B., et al. (2023). Veg-W2TCN: A parallel hybrid forecasting framework for non-stationary time series using wavelet and temporal convolution network model. *Applied Soft Computing*, 137, Article 110172.
- Sasal, L., Chakraborty, T., & Hadid, A. (2022). W-transformers: a wavelet-based transformer framework for univariate time series forecasting[C]. In *2022 21st IEEE international conference on machine learning and applications (ICMLA)* (pp. 671–676). IEEE.
- Song, B., Yu, Y., Zhou, Y., et al. (2018). Host load prediction with long short-term memory in cloud computing. *The Journal of Supercomputing*, 74, 6554–6568.
- Tanaka, G., Matsumori, T., Yoshida, H., et al. (2022). Reservoir computing with diverse timescales for prediction of multiscale dynamics. *Physical Review Research*, 4(3), Article L032014.
- Taylor, S J, & Letham, B. (2018). Forecasting at scale. *The American Statistician*, 72(1), 37–45.
- Teng, X., Zhang, L., Gao, P., et al. (2025). BERT-driven stock price trend prediction utilizing tokenized stock data and multi-step optimization approach. *Applied Soft Computing*, 170, Article 112627.
- Vaswani, A. (2017). *Attention is all you need*. In .
- Wang, H., Peng, J., Huang, F., et al. (2023). Micn: multi-scale local and global context modeling for long-term series forecasting[C]. In *The eleventh international conference on learning representations*.
- Wang S, Wu H, Shi X, et al. Timemixer: decomposable multiscale mixing for time series forecasting. arXiv preprint arXiv:2405.14616, 2024.
- Wei, W., Wang, Z., Pang, B., et al. (2025). Wavelet transformer: an effective method on multiple periodic decomposition for time series forecasting. *IEEE Transactions on Neural Networks and Learning Systems*.
- Wu, H., Xu, J., Wang, J., et al. (2021). Autoformer: decomposition transformers with auto-correlation for long-term series forecasting. *Advances in neural information processing systems*, 34, 22419–22430.
- Yang, B., Li, Y., Zheng, W., et al. (2023). Motion prediction for beating heart surgery with GRU. *Biomedical Signal Processing and Control*, 83, Article 104641.
- Yingwei, WANG, & Shucai, M. A. (2021). Time series forecasting based on hybrid ARIMA and LSTM models. *Computer Applications and Software*, 38(2), 291–298.
- Zeng, A., Chen, M., Zhang, L., et al. (2023). Are transformers effective for time series forecasting? [C]. In , 37. *Proceedings of the AAAI conference on artificial intelligence* (pp. 11121–11128).
- Zhao, Y., Norouzi, H., Azarderaksh, M., et al. (2021). Global patterns of hottest, coldest, and extreme diurnal variability on earth. *Bulletin of the American Meteorological Society*, 102(9), E1672–E1681.
- Zhao Y, Li T, Chen J, et al. Disentangled interpretable representation for efficient long-term time series forecasting. arXiv preprint arXiv:2411.17257, 2024.
- Zhao G, Zhou L, Wang X. AverageLinear: enhance long-term time series forecasting with simple averaging[J]. arXiv e-prints, 2024: arXiv: 2412.20727.
- Zhou, H., Zhang, S., Peng, J., et al. (2021). Informer: beyond efficient transformer for long sequence time-series forecasting[C]. In , 35. *Proceedings of the AAAI conference on artificial intelligence* (pp. 11106–11115).
- Zhou, T., Ma, Z., Wen, Q., et al. (2022). Fedformer: frequency enhanced decomposed transformer for long-term series forecasting[C]. In *International conference on machine learning* (pp. 27268–27286). PMLR.
- Zhu, Y., Zhang, W., Chen, Y., et al. (2019). A novel approach to workload prediction using attention-based LSTM encoder-decoder network in cloud environment. *EURASIP Journal on Wireless Communications and Networking*, 2019, 1–18.